

Experiment No. 17

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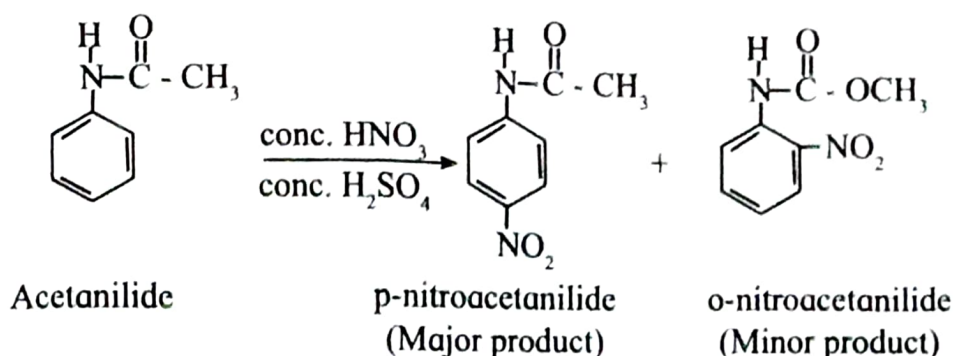
Aim: To prepare a pure sample of p-nitroacetanilide from acetanilide.

Theory: p-nitroacetanilide is prepared by the nitration of acetanilide by using nitrating mixture (conc. HNO_3 and conc. H_2SO_4) as nitrating reagent. The mixture of these two acids releases nitronium ion (NO_2^+), which acts as an electrophile in the reaction.



Nitronium ion attacks the benzene ring containing anilide group, mainly at the para position to give p-nitroacetanilide as a major product.

This is an example of aromatic electrophilic substitution reaction



Apparatus: Test tubes, conical flask, glass rod, funnel, measuring cylinder, beaker, etc.

Chemicals: conc. HNO_3 , conc. H_2SO_4 , glacial acetic acid, acetanilide, ethanol, ice.

Procedure:

1. In a 100 mL beaker dissolve 2 g of powdered acetanilide, add 2 mL of glacial acetic acid.
2. Add 4 mL of conc. H_2SO_4 gradually with stirring to the above mixture. The mixture becomes hot and clear solution is obtained. Cool the reaction mixture in an ice bath maintaining temperature $0-5^\circ\text{C}$.
3. Add a cold mixture of 1 mL conc. HNO_3 and 1 mL conc. H_2SO_4 to the viscous reaction mixture drop wise with constant stirring, so that the temperature of the mixture does not rise above 10°C .
4. Remove the beaker from the ice bath and allow the reaction mixture to attain room temperature. Stir the reaction mixture continuously and pour it on approximate 100 g of crushed ice.
5. Stir the mixture well and filter the compound obtained.
6. Wash the compound with cold water and dry it.
7. Recrystallize a small amount of the pale yellow solid by alcohol. Colourless crystals of p-nitroacetanilide are obtained. Yellow o-nitroacetanilide formed in the small amount remains in the mother liquor.
8. Dry and record the yield.

Observation and Result:

1. Yield of p-nitroacetanilide crystals =1.8.....g
2. Colour of p-nitroacetanilide crystals = ..colourless..

Remark and sign of teacher:

MCQ

Select the correct answer for the following multiple choice type questions.

- The +vely charged reagent used in the preparation of p-nitroacetanilide is
 a. nitronium ion ☒ b. hydronium ion
 c. nitrosonium ion d. hydrogen ion
- m-nitroacetanilide can NOT be formed during nitration of acetanilide, because
☒ a. -NHCOCH_3 is meta directing b. -NHCOCH_3 is para directing
 c. -NHCOCH_3 is ortho directing d. -NHCOCH_3 is ortho and para directing
- During nitration, in a nitrating mixture, conc. H_2SO_4 acts as
☒ a. dehydrating agent b. electrophile c. nucleophile d. hydrating agent
- Nitration of acetanilide is carried out at
☒ a. less than 5°C b. more than 15°C c. less than 100°C d. more than 100°C
- Freezing mixture is
 a. ice + common salt b. ice + cane sugar
 c. ice + common sugar ☒ d. only ice

Short answer questions

1. In the preparation of p-nitroacetanilide another minor product is formed. Write the name of compound and how it can be separated from p-nitroacetanilide?

Ans. In preparation of p-nitroacetanilide, o-nitroacetanilide is formed as minor product. It is formed with the mixture is dissolved in ethanol, more soluble o-nitroacetanilide.

2. What is meant by nitrating mixture?

Ans. Equimolar mixture of conc. HNO_3 & conc. H_2SO_4 .

3. Name the electrophile generated during the preparation of p-nitroacetanilide reaction.

Ans. Nitronium ion (NO_2^+) acts as electrophile in this reaction.

4. Write the colour of minor product formed in the preparation of p-nitroacetanilide.

Ans. Yellow colour o-nitroacetanilide is minor product formed in preparation of p-nitroacetanilide.

5. Write the structures of functional group present in p-nitroacetanilide.

Ans. Nitro group ($-\text{NO}_2$), amide ($\text{CH}_3-\text{C}(=\text{O})-\text{NH}-$)

Remark and sign of teacher: